Robot Arm

Carlos Ghost

VIEW IN BROWSER

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Summary

My first Robot Arm from home project :-)

Hobby & Makers > RC & Robotics

Tags: robot arm

Robot Arm uses the most common SG90 mini-servo motors and was powered by Arduino Nano.

Robot is very good helper in holding various small things (e.g. Soldering, Lights etc.).

Control for Robot can be with Joysticks or directly program routine.

Each servo have added a different step-down gear to increase the torque and reduce the speed to a reason manner.

Engineering Notes:

The waist servo have a speed reduction of 1:2 and 90 degree of rotation.

The upper arm servo have a speed reduction of 1:3 and 60 degree of rotation.

The lower arm servo have a speed reduction of 1:4 and 45 degree of rotation.

The base turning servo have a speed reduction of 1:5 and had extended the servo 5K VR from the servo in order to provide sufficient torque and 180 degree angle of rotation.

Six 4mm dia metal ball are required to support the base turning gear.

All servo motors are using the same gear of 9 teeth. This gear of 6 different tolerance/clearances are provided.

Material

- 5x Servo SG90
- 1x Hex Socket Head Cap Screw M2x10 DIN 912
- 1x Button Head Screw Flanged M2.5x25 ISO 7380-2
- 3x Hex Socket Head Cap Screw M2.5x20 DIN 912
- 1x Hex Socket Head Cap Screw M2.5x15 DIN 912
- 8x Hex Socket Head Cap Screw M2.5x8 DIN 912
- 2x Hex Socket Head Cap Screw M3x10 DIN 912
- 1x Hex Nut M2 DIN 934
- 2x Hex Nut M2.5 DIN 934
- 4x Self Locking Nut M2.5 DIN982
- 2x countersunk head tapping screw 2.2x6 DIN 7982
- 1x countersunk head tapping screw 2.2x8 DIN 7982
- 6x metal balls with 4mm diameter

Program for Arduino in pictures.

Functions for Robto Arm:

1. Manipulation

GHOST Robot Arm - YouTube

Axis on Robot Arm (youtube.com)

2. Holder for Soldering

Holder for Robot Arm (youtube.com)

3. Holder for Light

Light for Robot Arm (youtube.com)

Joystick - Joystick for Robot Arm od autora Carlos Ghost | Stáhněte si zdarma STL model | Printables.com

Holder - Holder od autora Carlos Ghost | Stáhněte si zdarma STL model | Printables.com

Box for Arduino and Battery - Box for Arduino od autora Carlos Ghost | Stáhněte si zdarma STL model | Printables.com

Enjoy!

Model files

7-arm-l_leg_3pcs.stl

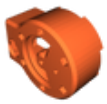
8-arm-l_middle.stl

11-arm-m_hsg1.stl

12-arm-m_hsg2.stl

15-arm-s_hsg2.stl

16-arm-s_middle.stl



1-arm-l_base.stl

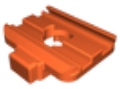
9-arm-l_nutknob.stl

13-arm-m_middle.stl

2-arm-l_basetopgear.stl

10b-arm-l_vrmount_v3mod.stl

21-lowerbearing.stl



18-clamp_main.stl

6-arm-l_hsg2.stl

14-arm-s_hsg1.stl

19-clamp_ymount.stl

5-arm-l_hsg1.stl

4arm-l_gearvr.stl

10-arm-l_vrmount_v1mod.stl

3-arm-l_gearcenter.stl

17-clamp_jaw_2pcs.stl

10a-arm-l_vrmount_v2mod.stl

20-gear1s_5pcs.stl

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